

# MATRICES

## The Art of Structured Data

LINEAR EQUATIONS | CRYPTOGRAPHY | SPREADSHEET LOGIC



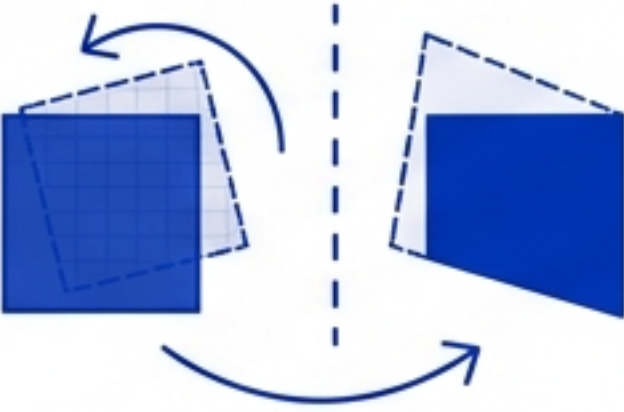
# A Tool for Complexity

The evolution of matrices began as a compact method to solve linear equations. Today, they are the engine behind modern data processing.

### Spreadsheets & Logic

|   | A | B |
|---|---|---|
| 1 |   |   |
| 2 |   |   |

### Transformation



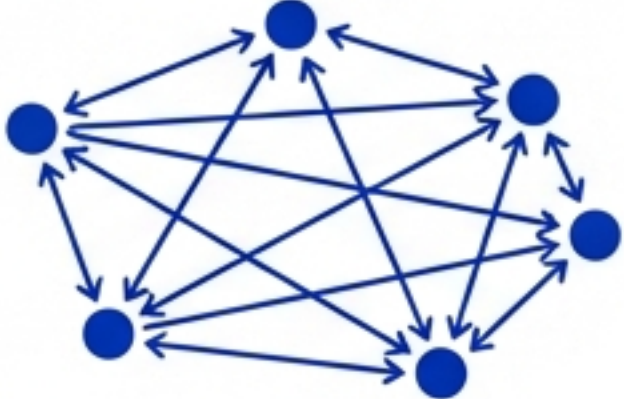
### Cryptography

59G3\$@#1  
\*%&+=', c  
56\{\$@0#  
>74\_→q#?

⇒



### Complex Systems





# From Chaos to Order

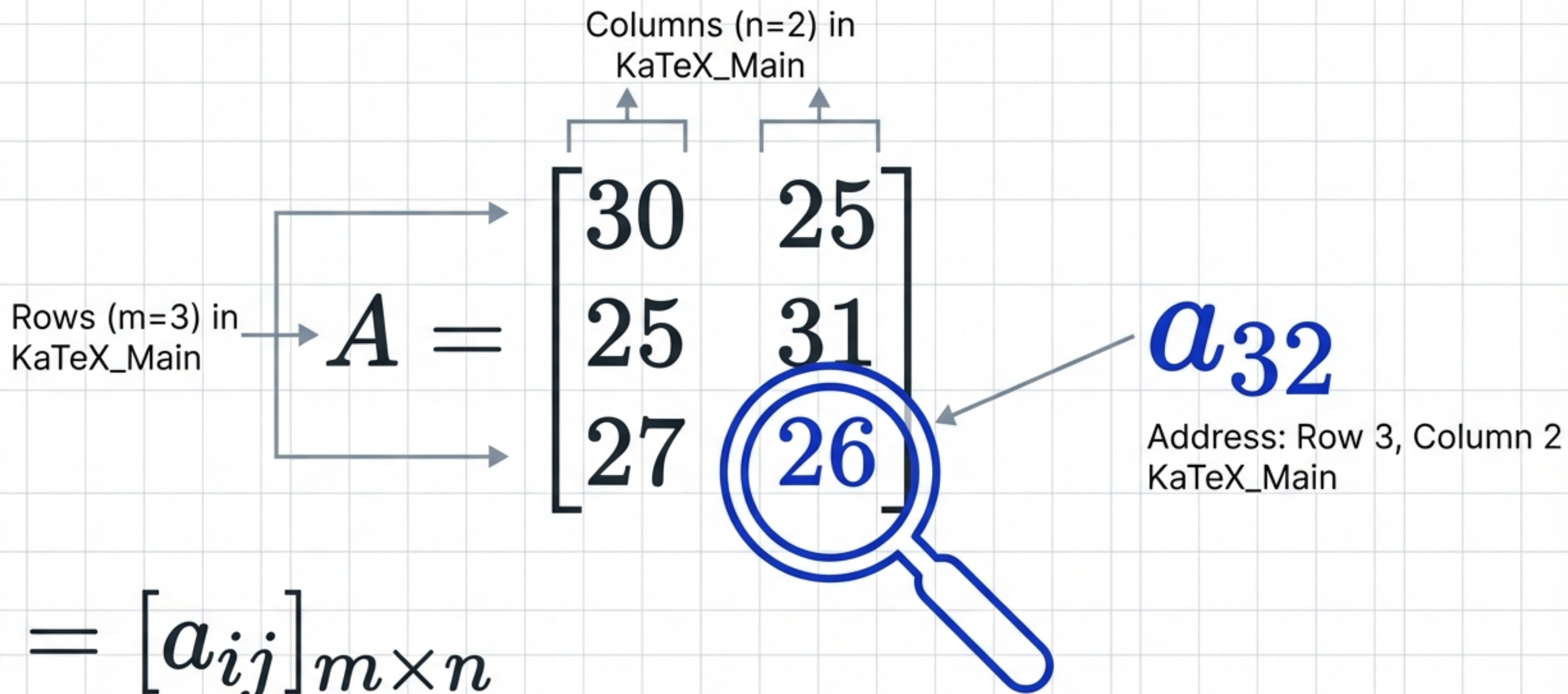
Radha has 15 notebooks and 6 pens. Fauzia has 10 notebooks and 2 pens. Simran has 13 notebooks and 5 pens.

|        | Notebooks | Pens |
|--------|-----------|------|
| Radha  | 15        | 6    |
| Fauzia | 10        | 2    |
| Simran | 13        | 5    |

$$\begin{bmatrix} 15 & 6 \\ 10 & 2 \\ 13 & 5 \end{bmatrix}$$

**Definition:** An ordered rectangular array of numbers.

# Anatomy of a Matrix





# Classifying by Shape

$$\begin{bmatrix} 0 \\ \sqrt{3} \\ -1 \\ \frac{1}{2} \end{bmatrix}$$

$m = 4$   
 $n = 1$

Column Matrix  
Order  $4 \times 1$

$$\left[ -\frac{1}{2} \quad \sqrt{5} \quad 2 \quad 3 \right]$$

$m = 1$   
 $n = 4$

Row Matrix  
Order  $1 \times 4$

$$\begin{bmatrix} 3 & -1 & 0 \\ \frac{3}{2} & 3\sqrt{2} & 1 \\ 4 & 3 & -1 \end{bmatrix}$$

$m = 3$   
 $n = 3$

Square Matrix  
Order  $3 \times 3$  ( $m = n$ )

# The Diagonal Hierarchy

Diagonal Matrix  $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$  Non-diagonal elements are zero.

↓ More Specific

Scalar Matrix  $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$  Diagonal elements are equal ( $k$ ).

↓ More Specific

Identity Matrix ( $I$ )  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  Diagonal elements are 1.

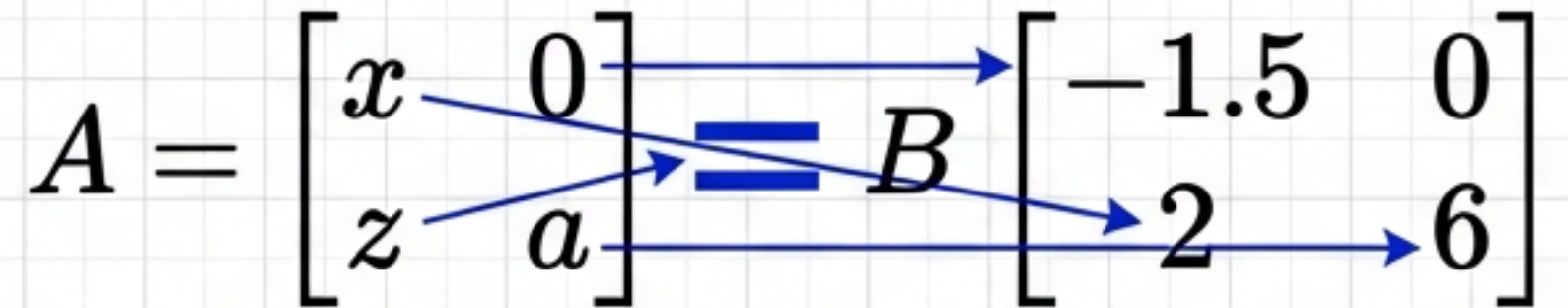


# The Rule of Equality

## Condition

- 1. Same Order.
- 2. Corresponding elements are equal ( $a_{ij} = b_{ij}$ ).

## Application

$$A = \begin{bmatrix} x & 0 \\ z & a \end{bmatrix} = B = \begin{bmatrix} -1.5 & 0 \\ 2 & 6 \end{bmatrix}$$


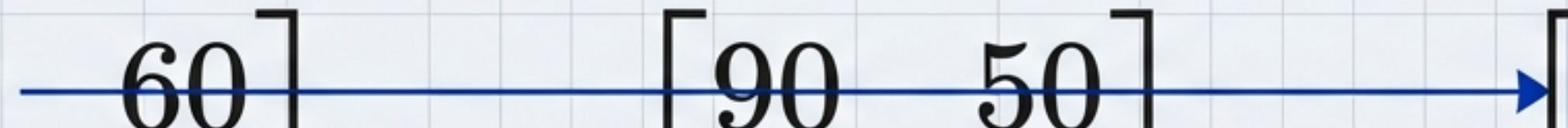
$$x = -1.5, z = 2, a = 6$$

$$C = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix} \neq D = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$$



# Operations: Adding Data

| Factory A                                                     |     | Factory B                                                     |     | Result Matrix                                                       |
|---------------------------------------------------------------|-----|---------------------------------------------------------------|-----|---------------------------------------------------------------------|
| $\begin{bmatrix} 80 & 60 \\ 75 & 65 \\ 90 & 85 \end{bmatrix}$ | $+$ | $\begin{bmatrix} 90 & 50 \\ 70 & 55 \\ 75 & 75 \end{bmatrix}$ | $=$ | $\begin{bmatrix} 170 & 110 \\ 145 & 120 \\ 165 & 160 \end{bmatrix}$ |

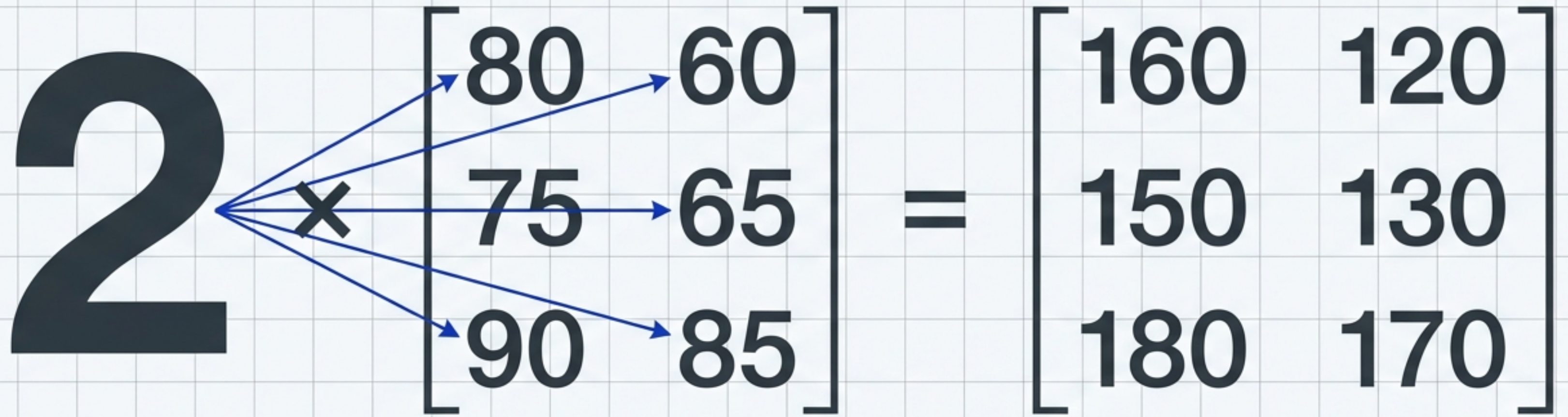


Requirement: Matrices must be of the same order.



# Operations: Scalar Multiplication

Doubling Production



The diagram illustrates the scalar multiplication of a 2x2 matrix by the scalar 2. On the left, a large number '2' is followed by a multiplication symbol 'x'. To the right of the 'x' is a 2x2 matrix with elements 80, 60, 75, and 85. Blue arrows point from the '2' to each element of the matrix, indicating the scalar is applied to each. To the right of the matrix is an equals sign, followed by another 2x2 matrix with elements 160, 120, 150, and 170. This shows the result of doubling each element of the original matrix.

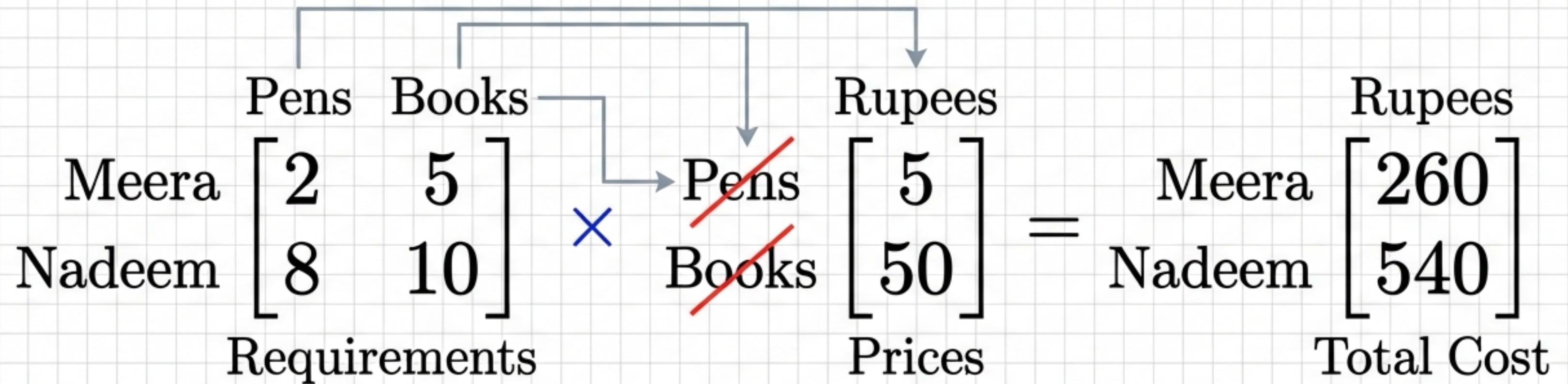
$$2 \times \begin{bmatrix} 80 & 60 \\ 75 & 85 \end{bmatrix} = \begin{bmatrix} 160 & 120 \\ 150 & 170 \end{bmatrix}$$

$$k \cdot A = [k(a_{ij})]$$



# Multiplication: The Logic

Meera and Nadeem shopping example to explain dimensions.



Sum of Products:  $(\text{Quantity} \times \text{Price}) + (\text{Quantity} \times \text{Price})$



# The Mechanism: Row Dives into Column

$$C = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 4 \end{bmatrix} \times D = \begin{bmatrix} 2 & 7 \\ -1 & 1 \\ 5 & -4 \end{bmatrix}$$

$$(1 \times 2) + (-1 \times -1) + (2 \times 5) \Rightarrow \begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix} = \begin{bmatrix} 13 & \dots \\ \dots & \dots \end{bmatrix}$$
$$= 2 + 1 + 10 = 13$$



# The Commutativity Trap

$A \times B$

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 4 \end{bmatrix} \times \begin{bmatrix} 2 & 7 \\ -1 & 1 \\ 5 & -4 \end{bmatrix}$$
$$\begin{bmatrix} -10 & -2 \\ 16 & -2 \end{bmatrix}$$

$\neq$

$B \times A$

$$\begin{bmatrix} 2 & 7 \\ -1 & 1 \\ 5 & -4 \end{bmatrix} \times \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 4 \end{bmatrix}$$
$$\begin{bmatrix} -10 & 2 & 21 \\ 16 & -2 & 37 \\ -2 & -2 & -11 \end{bmatrix}$$

Order matters. The product matrices are not even the same size.



# Zero Matrix Anomalies

$$\begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix} \times \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Non-Zero                      Non-Zero                      Zero Matrix (0)

Anomaly: The product of two non-zero matrices can be zero.



# Properties of Multiplication

## 1. Associative Law

$$(AB)C = A(BC)$$

Grouping doesn't change the result, as long as the order sequence is maintained.

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## 2. Distributive Law

$$A(B + C) = AB + AC$$

Multiplication distributes across addition.

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## 3. Multiplicative Identity

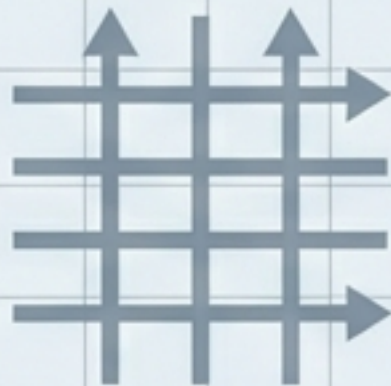
$$IA = AI = A$$

The Identity Matrix ( $I$ ) acts like the number 1.



# The Data Architect's Toolkit

## Structure



Order = Rows  $\times$  Columns

## Operations

- Add (same order)
- Scale (distribute)
- Multiply (Row-by-Column)

## Types

- Square ( $m = n$ )
- Diagonal
- Identity ( $I$ )
- Zero ( $O$ )

## The Watch Out

$$AB \neq BA$$



Matrices are not just representation; they are the machinery for transforming data.